

PHYS 143 – Mechanics and Thermal Physics

Newton's Laws and Forces

How do forces determine how an object moves?

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Reading: Young and Freedman, *University Physics* (15th ed.), Ch. 4, §4.1–4.6 and Ch. 5, §5.1–5.4

By the end of this week, you should be able to:

1. Identify forces and represent them using free-body diagrams.
2. Apply Newton's laws to equilibrium and accelerating systems.
3. Model normal force, tension and friction in physical systems.
4. Solve connected-object, inclined-plane and circular-dynamics problems.

The detailed outcomes are in the checklist at the end of the lecture.

Three contacts:

1 REPRESENT

What forces act, and how should we represent them?

2 ANALYSE

How do forces determine acceleration?

3 APPLY

How do contact forces and curved motion extend the model?

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. A 50 N force acts 30° above the $+x$ axis. Find its components.
2. A car goes from rest to 20 m/s in 8.0 s. Find its acceleration.
3. Forces of 3.0 N east and 4.0 N north act on a body. Find the magnitude and direction of the resultant.
4. A ball is thrown upward. What are its velocity and acceleration at the top?
5. Solve $T - 20 = 2a$ and $30 - T = 3a$ for T and a .
6. An object moves in a circle of radius 2.0 m at 4.0 m/s. Find its acceleration and its direction.

Check: brief review

1. $F_x = 50 \cos 30^\circ = 43.3 \text{ N}$,
 $F_y = 50 \sin 30^\circ = 25 \text{ N}$.
2. $a = 20/8.0 = 2.5 \text{ m/s}^2$.
3. $R = \sqrt{3^2 + 4^2} = 5.0 \text{ N}$ at
 $\tan^{-1}(4/3) = 53^\circ$ north of east.
4. $v = 0$ (momentarily), $a = 9.8 \text{ m/s}^2$
downward ($a_y = -g$).
5. Add: $10 = 5a$, so $a = 2.0 \text{ m/s}^2$ and
 $T = 24 \text{ N}$.
6. $a = v^2/R = 16/2.0 = 8.0 \text{ m/s}^2$, toward
the centre.

Calculus is not needed this week. The Week 1 and Week 2 toolkits (components, signs, circular acceleration) are reused throughout.

Contact 1: Represent

What forces act, and how should we represent them?

Weeks 1 and 2

vectors $\rightarrow \vec{r}(t) \rightarrow \vec{v}(t) \rightarrow \vec{a}(t)$

How does an object move?

Week 3

interaction $\rightarrow$ force $\rightarrow$ net force $\rightarrow \vec{a}$

Why does its motion change?

Opening question

A ball is thrown straight up. At its highest point, **which forces act on it, and what is the net force?**

Predict first

Poll: (a) none, (b) weight only, (c) weight and an upward “force of the throw”, (d) not enough information. Decide before the next slide.

top of flight: $v = 0$



Earth on ball

(b) Weight only.

- Only one object interacts with the ball: the Earth. The force is the **Earth on the ball**, its weight.
- There is no “force of the throw”: the hand stopped acting when the ball left it.
- The net force is mg downward, so $a = g$ downward (Week 2), even though $v = 0$ for an instant.

A force is an interaction between two objects

Rule: naming a force

Every force is an interaction between **two** objects. Name it by who exerts it on whom:

$$\vec{F}_{A \text{ on } B} = \text{the force exerted by } A \text{ on } B$$

- A force is a **vector**: magnitude and direction. Unit: newton, $1 \text{ N} = 1 \text{ kg m/s}^2$.
- **Contact** forces: pushes, pulls, surfaces, ropes.
- **Long-range** forces: gravity (and electric and magnetic forces later).

Physical meaning

If you cannot say *which other object* exerts a force, it is not a force. This single habit will keep your free-body diagrams correct.

Four common forces, named by agent

Force	Symbol	Agent on object	Direction
weight	$\vec{w}$	Earth on object	toward the Earth, $w = mg$
normal	$\vec{n}$	surface on object	perpendicular to the surface, away from it
tension	$\vec{T}$	rope on object	along the rope, pulling the object
friction	$\vec{f}$	surface on object	parallel to the surface, against the relative sliding
applied	$\vec{F}$	person or machine on object	as applied

At first, write the agent next to every force (“Earth on box”); later you can compress to w , n , T and f .

Rule: net force

Several forces on one object have the same effect as their **vector sum**:

$$\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \cdots \quad \sum F_x = F_{1x} + F_{2x} + \cdots, \quad \sum F_y = F_{1y} + F_{2y} + \cdots$$

This is Week 1 vector addition by components: **resolve, combine, reconstruct**.

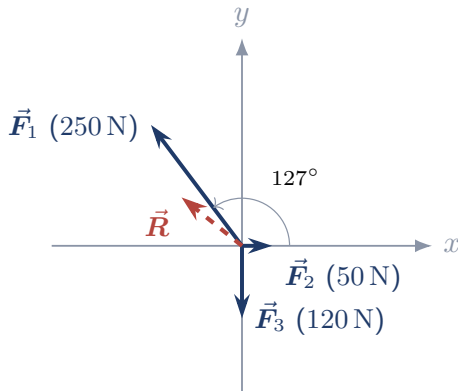
Lecturer example: net force of three forces

Question

Three forces act on a body: $F_1 = 250\text{ N}$ at 127° from $+x$, $F_2 = 50\text{ N}$ along $+x$, and $F_3 = 120\text{ N}$ along $-y$. Find the net force.

Predict first

From the sketch: which quadrant is the net force in?



Identify: target $\sum \vec{F}$. **Set up:** x to the right, y up; angles from $+x$.

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	magnitude	x -component	y -component
$\vec{F}_1$	250 N	$250 \cos 127^\circ = -150 \text{ N}$	$250 \sin 127^\circ = +200 \text{ N}$
$\vec{F}_2$	50 N	+50 N	0
$\vec{F}_3$	120 N	0	-120 N
$\sum \vec{F}$		-100 N	+80 N

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$\sum \vec{F}$		-100 N	+80 N

Execute: $R = \sqrt{100^2 + 80^2} = 128 \text{ N}$; reference angle $\tan^{-1}(80/100) = 38.7^\circ$ in quadrant II, so $\theta = 141^\circ$ from $+x$.

Evaluate: quadrant II, as predicted; F_1 dominates. Units N. ✓

A sliding puck

A hockey puck is pushed along a rough floor and stops. On smooth ice it slides much farther. What would happen on a perfectly smooth surface?

The puck slows down because the **floor exerts a friction force on it**, not because motion *needs* a force. With less friction, the velocity changes less. With none, it does not change.

Rule: Newton's first law

If the net force on an object is zero, its velocity is constant: it stays at rest, or moves in a straight line at constant speed.

$$\sum \vec{F} = 0 \Rightarrow \vec{a} = 0$$

Physical meaning

Inertia: the tendency to keep the same velocity. The law holds in an **inertial frame**, such as the ground or a train at constant velocity.

Concept check: is the net force zero?

In which cases is $\sum \vec{F} = 0$?

1. A ball at the top of its flight.
2. A puck gliding at constant speed on frictionless ice.
3. A passenger sitting in a train moving smoothly at constant velocity.
4. A car going round a bend at constant speed.

Predict first

Hint: the first law concerns the *velocity vector*, not only the speed.

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Predict first

Hint: the first law concerns the *velocity vector*, not only the speed.

Answer: (2) and (3). (1) $a = g$, so $\sum \vec{F} \neq 0$. (4) the direction of $\vec{v}$ changes, so $\vec{a} \neq 0$ and $\sum \vec{F} \neq 0$.

Common error: “no motion needed no force, and motion needs a force.” Motion needs no force; *changing* the velocity does.

- **Mass** m (kg): how much matter, and how strongly the object resists acceleration. The same everywhere.
- **Weight** w (N): the gravitational force of the Earth on the object.

Rule: weight

$$w = mg$$

directed toward the Earth's centre.

Example

A 2.0 kg object: weight on Earth ($g = 9.8 \text{ m/s}^2$) and on the Moon ($g = 1.62 \text{ m/s}^2$)?

Mass and weight

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Example

A 2.0 kg object: weight on Earth ($g = 9.8 \text{ m/s}^2$) and on the Moon ($g = 1.62 \text{ m/s}^2$)?

$w_{\text{Earth}} = 19.6 \text{ N}$, $w_{\text{Moon}} = 3.2 \text{ N}$; the mass is 2.0 kg in both places.

Free fall: $a = w/m = g$ for every mass, so a heavier object does not fall faster (Week 2).

TOOLKIT 1 – Free-body diagrams

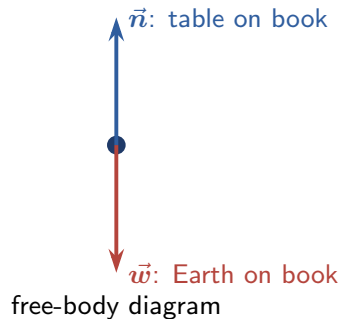
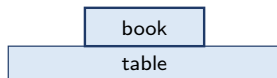
1. **ISOLATE** one body and draw it as a dot.
2. **ASK** “Which other objects interact with this body?” Everything touching it, and the Earth.
3. **DRAW** one arrow per interaction, starting at the dot, in the direction the force acts *on the body*. Label each with its agent: “Earth on box (w)”, “table on box (n)”, and so on.
4. **AXES** are added afterwards (Toolkit 2).

RULE: only forces acting *on* the body, each with a named agent.

WATCH OUT: do not draw the net force, ma , the velocity or forces the body exerts on other objects.

CHECK: can I name an agent for every arrow, and have I missed any object touching the body?

Build it together: a book on a table

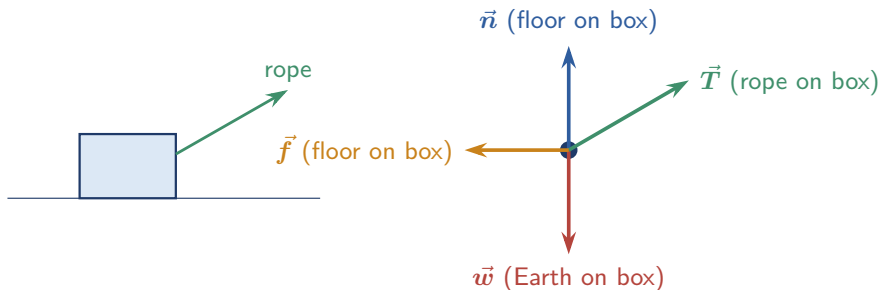


Predict first

If the book is at rest, how do n and w compare? Why?

$\sum \vec{F} = 0$ (first law), so $n = w$: the table pushes up exactly as hard as the Earth pulls down.

Build it together: a box pulled by a rope



Four interactions, four forces

Earth on box ($\vec{w}$), floor on box ($\vec{n}$ and $\vec{f}$), rope on box ($\vec{T}$). The rope's force is angled, so it will have components.

Your turn: draw the free-body diagrams

Your turn

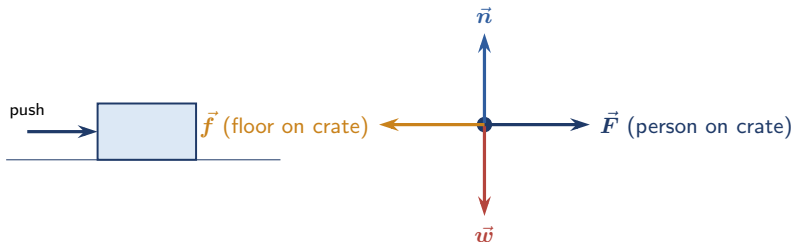
Draw a free-body diagram, with agent names, for each situation:

- (a) a crate pushed along a rough floor at **constant velocity**;
- (b) a block sliding down a **rough** incline.

Predict first

For (a): how do the push and the friction compare? For (b): in which direction does friction point?

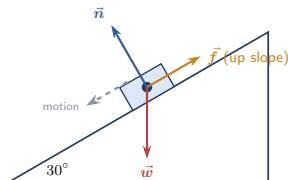
Your turn: answer (a)



(a) A crate pushed at constant velocity

Four forces: $\vec{w}$ (Earth on crate), $\vec{n}$ (floor on crate), $\vec{F}$ (person on crate), $\vec{f}$ (floor on crate). At constant velocity $\sum \vec{F} = 0$, so the push equals the friction, and $n = w$.

Your turn: answer (b)



(b) A block sliding down a rough incline

Three forces: $\vec{w}$, $\vec{n}$ (perpendicular to the surface) and $\vec{f}$ (up the slope, against the sliding).

Common error

There is no separate “force down the slope”: that is a component of the weight, which we use in Contact 3.

Contact 1: what we can now do

Interaction



Force (vector, named by agent)



Net force



First law: $\sum \vec{F} = 0 \Rightarrow \vec{a} = 0$

You should be able to

- name a force by who exerts it on whom;
- find a net force by components;
- state and apply the first law;
- distinguish mass from weight;
- draw a correct free-body diagram.

✓ Before Contact 2

If the net force is not zero, what happens? Draw FBDs for two everyday situations. Read Young and Freedman, §4.3 and §5.1.

Contact 2: Analyse

How do forces determine acceleration?

Poll

The same 20 N horizontal push is applied to a 40 kg box and to an 80 kg box on a smooth floor. How do the accelerations compare?

Predict first

Equal? Half? Double? And how would the acceleration change if the push were doubled?

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Predict first

Equal? Half? Double? And how would the acceleration change if the push were doubled?

The 80 kg box accelerates half as much. Doubling the force doubles the acceleration: $a \propto F$ and $a \propto 1/m$.

Rule: Newton's second law

$$\sum \vec{F} = m\vec{a} \quad \Longleftrightarrow \quad \sum F_x = ma_x, \quad \sum F_y = ma_y$$

The acceleration has the direction of the net force and is inversely proportional to the mass.

- $\sum \vec{F}$ means the net force: all forces *on this object*.
- Unit: $1 \text{ N} = 1 \text{ kg m/s}^2$.
- Special case: $\vec{a} = 0$ gives equilibrium (first law).

Model assumptions

Valid in an inertial frame, for an object treated as a particle with constant mass. The momentum form $\sum \vec{F} = d\vec{p}/dt$ comes in a later lecture.

TOOLKIT 2 – Choosing axes and resolving forces

CHOOSE axes to simplify the physics, not automatically horizontal and vertical. Put one axis along the acceleration (or the motion) and the other perpendicular to it. On an incline: x parallel to the surface, y perpendicular.

RESOLVE each force that is not along an axis into components, using the Week 1 toolkit: the angle from an axis, sketch the signs first.

SIGNS: a component is positive along the positive axis direction, negative against it.

WATCH OUT: a force with no component along an axis does not appear in that axis equation.

CHECK: do the components, recombined, give back the original force?

✂ TOOLKIT 3 – Applying $\sum \vec{F} = m\vec{a}$ by components

1. Draw the free-body diagram (Toolkit 1).
2. Choose the axes and resolve forces (Toolkit 2).
3. Write one equation per axis: $\sum F_x = ma_x$, $\sum F_y = ma_y$, *before* inserting numbers.
4. Count unknowns and equations; they must match.
5. Solve algebraically, then substitute with units. For equilibrium put $a = 0$.
6. Evaluate: sign, units, size, a limiting case.

WATCH OUT: n is found from the equation perpendicular to the surface; do not assume $n = mg$.

CHECK: does the direction of $\vec{a}$ agree with the direction of the net force in the diagram?

Question

A worker pushes a 40 kg box on a smooth floor with a constant horizontal force of 20 N. Find its acceleration.

Predict first

Sketch the FBD. Which forces appear in the x equation? Roughly: less than or more than 1 m/s^2 ?

Question

A worker pushes a 40 kg box on a smooth floor with a constant horizontal force of 20 N. Find its acceleration.

Predict first

Sketch the FBD. Which forces appear in the x equation? Roughly: less than or more than 1 m/s^2 ?

Set up: $+x$ along the push. The forces $\vec{w}$ and $\vec{n}$ are vertical and cancel. **Execute:**

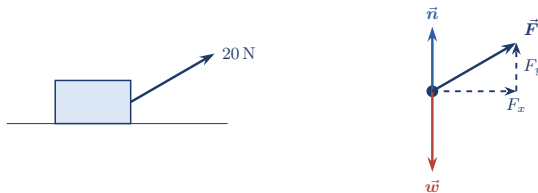
$$\sum F_x = 20 = 40 a_x \Rightarrow a_x = 0.50 \text{ m/s}^2.$$

Evaluate: positive (along the push), units m/s^2 , less than 1 m/s^2 . ✓

Lecturer example: a pulled box

Question

The same 40 kg box is now pulled by a rope with a 20 N force at 30° above the horizontal. The floor is still smooth. Find the acceleration and the normal force.



Predict first

Will a be larger or smaller than 0.50 m/s^2 ? Will n be larger or smaller than mg ?

A pulled box: solution

Set up: x horizontal along the motion, y up. Resolve $\vec{F}$: $F_x = 20 \cos 30^\circ = 17.3 \text{ N}$,
 $F_y = 20 \sin 30^\circ = 10.0 \text{ N}$.

A pulled box: solution

Set up: x horizontal along the motion, y up. Resolve $\vec{F}$: $F_x = 20 \cos 30^\circ = 17.3 \text{ N}$,
 $F_y = 20 \sin 30^\circ = 10.0 \text{ N}$.

Execute:

$$x: \sum F_x = F_x = ma_x \Rightarrow a_x = \frac{17.3}{40} = 0.43 \text{ m/s}^2$$

$$y: \sum F_y = n + F_y - mg = 0 \Rightarrow n = mg - F_y = 392 - 10.0 = 382 \text{ N}$$

($mg = 40 \times 9.8 = 392 \text{ N}$)

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($mg = 40 \times 9.8 = 392 \text{ N}$)

Evaluate: $a < 0.50$, because only part of the pull is horizontal; $n < mg$, because the rope lifts part of the weight. ✓

Common error

$n = mg$ only when no other force has a component perpendicular to the surface.

Equilibrium: the special case $a = 0$

Rule: equilibrium

An object at rest, or moving at constant velocity, has $\sum \vec{F} = 0$:

$$\sum F_x = 0 \quad \sum F_y = 0$$

Equilibrium needs the *same* method: free-body diagram, axes, one equation per axis. The only difference is that the right-hand side is zero.

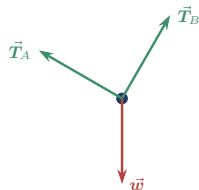
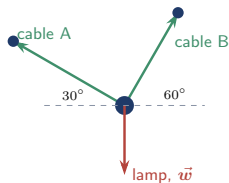
Predict first

A lamp hangs from two cables at different angles. Which cable carries more tension?

Lecturer example: a hanging lamp

Question

A 10.0 kg lamp hangs at rest from two cables. Cable A makes 30° with the horizontal and cable B makes 60° on the other side. Find the tensions.



free-body diagram of the lamp

Predict first

Which cable, A or B, carries more tension? Check this against the answer.

A hanging lamp: solution

Set up: $w = 10.0 \times 9.8 = 98.0 \text{ N}$. x to the right, y up. Components: T_A at 30° from $-x$, T_B at 60° from $+x$.

A hanging lamp: solution

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Execute:

$$x: -T_A \cos 30^\circ + T_B \cos 60^\circ = 0 \Rightarrow T_A = 0.577 T_B$$

$$y: T_A \sin 30^\circ + T_B \sin 60^\circ - 98.0 = 0$$

$$\text{Substitute: } (0.577)(0.500)T_B + 0.866 T_B = 98.0 \Rightarrow T_B = 84.9 \text{ N}, \quad T_A = 49.0 \text{ N}$$

A hanging lamp: solution

Set up: $w = 10.0 \times 9.8 = 98.0 \text{ N}$. x to the right, y up. Components: T_A at 30° from $-x$, T_B at 60° from $+x$.

Execute:

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Evaluate: cable B (steeper) carries more, as predicted. Vertical check:

$$49.0 \sin 30^\circ + 84.9 \sin 60^\circ = 24.5 + 73.5 = 98.0 \text{ N}. \quad \checkmark$$

The tensions are not both $w/2$: unequal angles give unequal tensions.

Your turn: a car accelerates

Your turn

A 1500 kg car speeds up from 10.0 m/s to 25.0 m/s in 6.0 s along a straight road. The resistive force on it is 400 N. Find the driving force exerted on the car by the road.

Predict first

Sketch the FBD. Is the driving force larger or smaller than the net force? Why?

Your turn

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Predict first

Sketch the FBD. Is the driving force larger or smaller than the net force? Why?

Week 2 kinematics: $a = (25.0 - 10.0)/6.0 = 2.5 \text{ m/s}^2$. Week 3 dynamics:

$\sum F_x = F - 400 = ma = 1500(2.5) = 3750 \text{ N}$, so $F = 4150 \text{ N}$.

Evaluate: larger than the net force, as predicted, because the resistive force must also be overcome. ✓

Rule: Newton's third law

If object A exerts a force on object B, then B exerts a force on A that is equal in magnitude and opposite in direction:

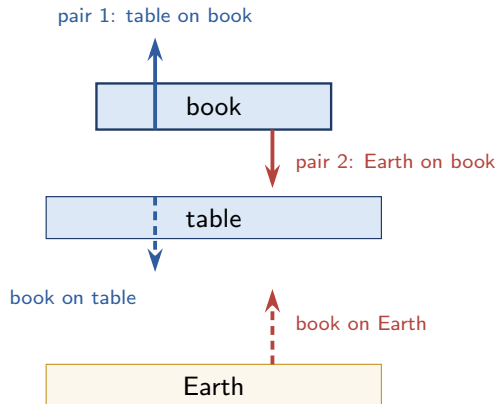
$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$$

- The two forces are the **same type** and act at the same time.
- They act on **different bodies**, so they never appear in the same free-body diagram, and they never cancel.

TOOLKIT 4 – Identifying third-law pairs

1. **NAME** the force: “A on B”. 2. **FLIP** it: the partner is “B on A”, the same type of force.
 3. **LOCATE** the partner: it acts on the *other* object, so it appears in that object’s free-body diagram, not in yours.
 4. **COMPARE** sizes: equal magnitude, opposite direction. The accelerations differ if the masses differ.
- RULE:** a pair is never two forces on the same object.
- WATCH OUT:** two forces that merely have equal size in a diagram are not a pair unless they are “A on B” and “B on A”.
- CHECK:** does each member of the pair act on a different object?

Concept check: pairs for the book on the table



Question

For the book at rest, are $\vec{w}$ (Earth on book) and $\vec{n}$ (table on book) a third-law pair?

Predict first

Apply Toolkit 4: what is the partner of $\vec{n}$? Of $\vec{w}$?

No. Both $\vec{w}$ and $\vec{n}$ act on the *book*.

- Partner of $\vec{n}$ (table on book): the force of the *book on the table*.
- Partner of $\vec{w}$ (Earth on book): the force of the *book on the Earth*.

Why are n and w equal here?

Because $\sum \vec{F} = 0$ on the book (first law), not because of the third law. If the table were in an accelerating lift, n and w would differ, but the pairs would still be equal and opposite.

Question

A horse pulls a cart forward. By the third law, the cart pulls back on the horse with an equal force. **Why can the horse and cart accelerate forward?**

Predict first

Draw the free-body diagram of the cart, and then of the horse, separately.

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Predict first

Draw the free-body diagram of the cart, and then of the horse, separately.

Cart: the forward pull by the horse is greater than the resistive forces, so it accelerates. **Horse:** the forward force on its hooves from the *ground* is larger than the backward pull from the cart. The pair (horse on cart, cart on horse) acts on different bodies and does not cancel in either diagram.

A heavy truck and a small car in a collision exert equal forces on each other; the car accelerates more because $a = F/m$.

Rule: ideal rope and pulley

An **ideal rope** is light and does not stretch: it has the same tension throughout, and objects joined by a taut rope have the same acceleration magnitude. An **ideal pulley** is light and frictionless and only changes the direction of the rope.

Physical meaning

Real ropes have mass and stretch, and real pulleys have friction. The ideal model is valid when these effects are small.

Common error

The tension is not automatically equal to a weight. It depends on the acceleration.

TOOLKIT 5 – Connected objects

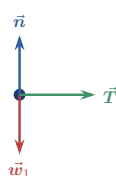
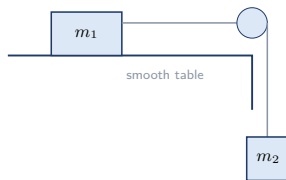
1. **ISOLATE** each body and draw its own free-body diagram.
2. **CONSTRAINT**: state the link: same acceleration magnitude, same rope tension (ideal rope and pulley).
3. **AXES**: choose each body's axis along its own motion, with signs consistent with the constraint.
4. **WRITE** one $\sum F = ma$ equation per body along its motion.
5. **SOLVE** the simultaneous equations for a and T .

WATCH OUT: drawing one diagram for the whole system and forgetting that internal tensions appear only when you separate the bodies.

CHECK: treat the whole system once: the external force equals the total mass times a . Check limiting cases.

Lecturer example: a block, a pulley and a hanging mass

A block $m_1 = 3.0 \text{ kg}$ on a smooth table is joined by an ideal rope over an ideal pulley to a hanging block $m_2 = 2.0 \text{ kg}$. Released from rest, find the acceleration and the tension.



FBD of m_1



FBD of m_2

Predict first

Is T larger or smaller than $m_2g = 19.6 \text{ N}$? Is a larger or smaller than g ?

Block, pulley and hanging mass: solution

Set up: m_1 : $+x$ toward the pulley; m_2 : $+y$ downward (consistent with the constraint). The same a and T for both.

Block, pulley and hanging mass: solution

Set up: m_1 : $+x$ toward the pulley; m_2 : $+y$ downward (consistent with the constraint). The same a and T for both.

Execute:

$$m_1 (x): T = m_1 a \quad m_2 (y, \text{down}): m_2 g - T = m_2 a$$

$$\text{Add: } m_2 g = (m_1 + m_2) a \Rightarrow a = \frac{m_2 g}{m_1 + m_2} = \frac{19.6}{5.0} = 3.9 \text{ m/s}^2; \quad T = m_1 a = 11.8 \text{ N}$$

Block, pulley and hanging mass: solution

Set up: m_1 : $+x$ toward the pulley; m_2 : $+y$ downward (consistent with the constraint). The same a and T for both.

Execute:

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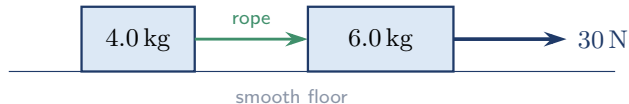
Evaluate: $T < m_2 g$ and $a < g$, as predicted. Whole-system check: the only external horizontal driving force is $m_2 g$, so $m_2 g = (m_1 + m_2) a$. ✓

Limits: $m_1 \rightarrow 0$ gives $a = g$, $T = 0$ (free fall); $m_2 \rightarrow 0$ gives $a = 0$.

Your turn: two blocks and a rope

Your turn

Two blocks on a smooth floor are joined by an ideal rope and pulled by a horizontal 30 N force as shown. Find the acceleration and the tension in the rope.



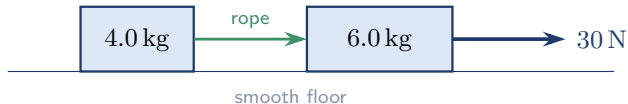
Predict first

Is T larger or smaller than 30 N? Draw both free-body diagrams.

Your turn: two blocks and a rope

Your turn

Two blocks on a smooth floor are joined by an ideal rope and pulled by a horizontal 30 N force as shown. Find the acceleration and the tension in the rope.



Predict first

Is T larger or smaller than 30 N? Draw both free-body diagrams.

Whole system: $a = \frac{30}{4.0 + 6.0} = 3.0 \text{ m/s}^2$. The 4.0 kg block: $T = m_1 a = 12 \text{ N}$. Check on the 6.0 kg block: $30 - 12 = 18 = 6.0(3.0)$. ✓ $T < 30 \text{ N}$, as predicted.

Contact 2: what we can now do

FBD



Axes and components



$$\sum F_x = ma_x, \quad \sum F_y = ma_y$$



Solve and evaluate

You should be able to

- apply the second law by components;
- solve equilibrium problems;
- identify third-law pairs and explain why they do not cancel;
- solve two-body connected problems.

✓ Before Contact 3

Read Young and Freedman, §5.3–5.4. Recall circular acceleration $a = v^2/R$ from Week 2: what force produces it?

Contact 3: Apply

How do contact forces and curved motion extend the model?

An opening puzzle: pushing a heavy crate

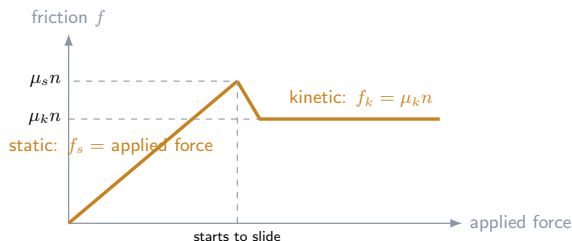
Poll

You push a heavy crate. Nothing happens. You push harder: still nothing. Harder again, and it suddenly slides. How does the floor's friction force change as you push harder?

Predict first

Sketch friction against applied force before the next slide.

Static and kinetic friction



Rule: friction

Static (no sliding):

$$0 \leq f_s \leq \mu_s n$$

Kinetic (sliding): $f_k = \mu_k n$

Direction: parallel to the surface, **against the relative sliding** (or the tendency to slide) of the two surfaces.

Usually $\mu_k < \mu_s$.

Strongest misconception of the week

$f_s = \mu_s n$ is the maximum static friction, not its value

Static friction adjusts itself to match the other forces, up to $\mu_s n$.

Example: a 40 kg crate on a floor with $\mu_s = 0.50$. The maximum static friction is $0.50 \times 392 = 196 \text{ N}$.

If you push with 150 N and the crate does not move, then $f_s = 150 \text{ N}$, *not* 196 N.

Two ways to get it wrong

- (1) Using $f_s = \mu_s n$ when the object is not about to slide.
- (2) Using $\mu_s n$ when the object is already sliding: then use $f_k = \mu_k n$.

✂ TOOLKIT 6 – Friction and the normal force

1. **NORMAL FORCE:** use the equation perpendicular to the surface to find n when n is not already known; do not assume $n = mg$.
2. **IS IT SLIDING?** If not, friction is whatever is needed (static), and must satisfy $f_s \leq \mu_s n$. Test with $\sum F = 0$.
3. **IF SLIDING,** $f_k = \mu_k n$, directed against the relative sliding of the surfaces.
4. Write $\sum F = ma$ along the motion (Toolkit 3).

WATCH OUT: friction does not always oppose motion relative to the ground; it opposes sliding relative to the *surface*.

CHECK: is $f_s \leq \mu_s n$ satisfied if the object is at rest?

Question

A 40.0 kg crate rests on a level floor with $\mu_s = 0.50$ and $\mu_k = 0.40$. (a) A horizontal push of 150 N is applied: does the crate move, and what is the friction? (b) The push is increased to 250 N: find the acceleration.

Predict first

Sketch the FBD. Estimate: which push is likely to exceed the maximum static friction?

Pushing a crate: solution

Set up: level floor, no vertical acceleration, so $n = mg = 40.0 \times 9.8 = 392 \text{ N}$. Maximum static friction: $\mu_s n = 0.50 \times 392 = 196 \text{ N}$.

Set up: level floor, no vertical acceleration, so $n = mg = 40.0 \times 9.8 = 392 \text{ N}$. Maximum static friction: $\mu_s n = 0.50 \times 392 = 196 \text{ N}$.

(a) $150 < 196 \text{ N}$: the crate does **not** move, so $f_s = 150 \text{ N}$ (balancing the push).

Set up: level floor, no vertical acceleration, so $n = mg = 40.0 \times 9.8 = 392 \text{ N}$. Maximum static friction: $\mu_s n = 0.50 \times 392 = 196 \text{ N}$.

(a) $150 < 196 \text{ N}$: the crate does **not** move, so $f_s = 150 \text{ N}$ (balancing the push).

(b) $250 > 196 \text{ N}$: it slides. $f_k = \mu_k n = 0.40 \times 392 = 157 \text{ N}$; $\sum F_x = 250 - 157 = 93 \text{ N}$;
 $a = 93/40.0 = 2.3 \text{ m/s}^2$.

Set up: level floor, no vertical acceleration, so $n = mg = 40.0 \times 9.8 = 392 \text{ N}$. Maximum static friction: $\mu_s n = 0.50 \times 392 = 196 \text{ N}$.

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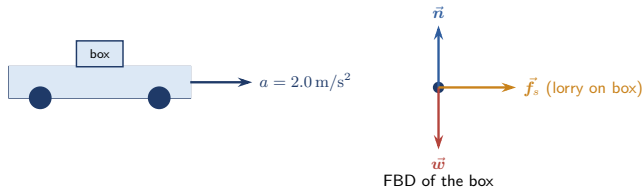
(b) $250 > 196 \text{ N}$: it slides. $f_k = \mu_k n = 0.40 \times 392 = 157 \text{ N}$; $\sum F_x = 250 - 157 = 93 \text{ N}$;
 $a = 93/40.0 = 2.3 \text{ m/s}^2$.

Evaluate: (a) $f_s \leq \mu_s n$ ✓. (b) once the crate slides, the friction drops from 196 N to 157 N, so the net force is positive. ✓

Concept check: friction can accelerate an object

Question

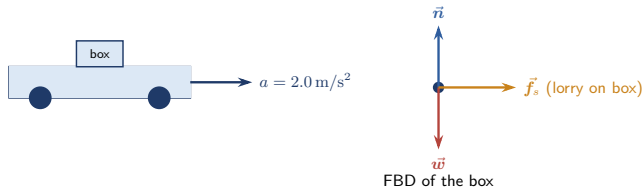
A box sits on the bed of a lorry that accelerates forward at 2.0 m/s^2 . The box does not slide. What force accelerates the box? What is the least μ_s ?



Concept check: friction can accelerate an object

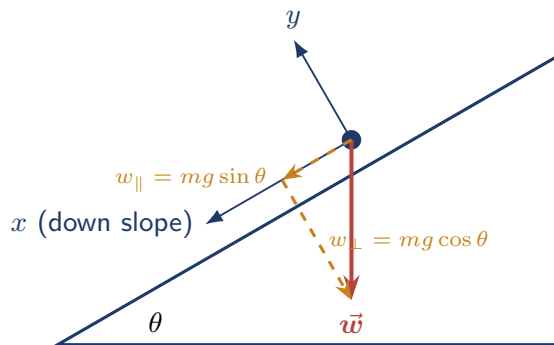
Question

A box sits on the bed of a lorry that accelerates forward at 2.0 m/s^2 . The box does not slide. What force accelerates the box? What is the least μ_s ?



The **static friction** of the lorry on the box points forward, along the box's motion: $f_s = ma$. It requires $\mu_s \geq a/g = 0.20$. Friction opposes the *sliding* between surfaces; it can point along the motion relative to the ground.

An inclined plane: choosing axes



Rule: weight on an incline

Axes: x along the slope, y perpendicular.

$$w_{\parallel} = mg \sin \theta$$

$$w_{\perp} = mg \cos \theta$$

Limits: $\theta = 0$ gives $w_{\parallel} = 0$; $\theta = 90^\circ$ gives $w_{\perp} = 0$. With no other perpendicular force, $n = mg \cos \theta$.

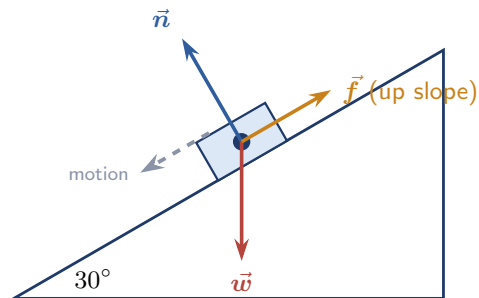
Lecturer example: sliding down an incline

Question

A 5.0 kg block slides down a 30.0° incline with kinetic friction $\mu_k = 0.30$. Find its acceleration.

Predict first

Is a larger or smaller than $g \sin 30^\circ = 4.9 \text{ m/s}^2$? Does the answer depend on the mass?



Sliding down an incline: solution

Set up: x down the slope (the direction of motion), y perpendicular. Forces: $\vec{w}$ (components $mg \sin \theta$ along $+x$, $mg \cos \theta$ along $-y$), $\vec{n}$ along $+y$, $\vec{f}$ along $-x$.

Sliding down an incline: solution

Set up: x down the slope (the direction of motion), y perpendicular. Forces: $\vec{w}$ (components $mg \sin \theta$ along $+x$, $mg \cos \theta$ along $-y$), $\vec{n}$ along $+y$, $\vec{f}$ along $-x$.

Execute:

$$y: n - mg \cos \theta = 0 \Rightarrow n = 5.0(9.8) \cos 30^\circ = 42.4 \text{ N}$$

$$f_k = \mu_k n = 0.30 \times 42.4 = 12.7 \text{ N}$$

$$x: mg \sin \theta - f_k = ma \Rightarrow 24.5 - 12.7 = 5.0 a \Rightarrow a = 2.4 \text{ m/s}^2$$

Sliding down an incline: solution

Set up: x down the slope (the direction of motion), y perpendicular. Forces: $\vec{w}$ (components $mg \sin \theta$ along $+x$, $mg \cos \theta$ along $-y$), $\vec{n}$ along $+y$, $\vec{f}$ along $-x$.

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$$x: mg \sin \theta - f_k = ma \Rightarrow 24.5 - 12.7 = 5.0 a \Rightarrow a = 2.4 \text{ m/s}^2$$

Evaluate: $a < 4.9 \text{ m/s}^2$, as predicted. Algebra: $a = g(\sin \theta - \mu_k \cos \theta)$, so the mass cancels.

With $\mu_k = 0$ this gives $g \sin \theta$. ✓

An opening question: a car on a bend

Poll

A car goes round a flat circular bend at a **constant speed**. Is the net force on it zero? If not, in which direction does it point?

Predict first

Recall from Week 2: what is the direction and size of the acceleration in uniform circular motion?

Rule: net force in uniform circular motion

The acceleration is $a_{\text{rad}} = v^2/R$ toward the centre (Week 2), so the net force must point toward the centre:

$$\sum F_{\text{radial}} = m \frac{v^2}{R} = m\omega^2 R$$

“Centripetal force” is not a new kind of force

It is the name for the *net inward force*, supplied by real forces: friction, tension, the normal force, gravity. It never appears as an extra arrow on a free-body diagram.

TOOLKIT 7 – Circular dynamics

1. **FBD** of the object with real forces only.
2. **RADIAL AXIS:** positive toward the centre.
3. **WRITE** $\sum F_{\text{radial}} = mv^2/R$ (or $m\omega^2 R$), and the equation for the perpendicular direction (for a flat curve, $n = mg$).
4. **IDENTIFY** which real force supplies the inward net force.
5. **LIMIT:** the maximum speed occurs when the supplying force reaches its maximum value.

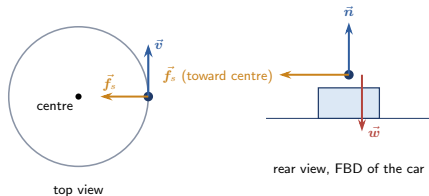
WATCH OUT: do not draw a “centrifugal” or “centripetal” force.

CHECK: does the net force point toward the centre, and is the size reasonable compared with mg ?

Lecturer example: a flat curve

Question

A 1200 kg car rounds a flat circular bend of radius 50.0 m at 15.0 m/s. (a) What friction force is required, and what is the least μ_s ? (b) What is the maximum speed for $\mu_s = 0.80$ (dry) and $\mu_s = 0.30$ (wet)?



Predict first

Which force points toward the centre? Does the answer depend on the mass?

A flat curve: solution

Set up: radial axis toward the centre; the only horizontal force is static friction, so f_s supplies the net inward force. Vertical: $n = mg = 11\,760\text{ N}$.

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$$(a) \quad f_s = \frac{mv^2}{R} = \frac{1200(15.0)^2}{50.0} = 5400\text{ N}; \quad \mu_{s,\min} = \frac{f_s}{n} = 0.46.$$

Set up: radial axis toward the centre; the only horizontal force is static friction, so f_s supplies the net inward force. Vertical: $n = mg = 11\,760\text{ N}$.

(a) $f_s = \frac{mv^2}{R} = \frac{1200(15.0)^2}{50.0} = 5400\text{ N}; \mu_{s,\min} = \frac{f_s}{n} = 0.46.$

(b) At the limit $f_s = \mu_s mg = mv^2/R$, so $v_{\max} = \sqrt{\mu_s g R}$ (the mass cancels).

Dry: $\sqrt{0.80 \times 9.8 \times 50.0} = 19.8\text{ m/s}$. Wet: $\sqrt{0.30 \times 9.8 \times 50.0} = 12.1\text{ m/s}$.

Set up: radial axis toward the centre; the only horizontal force is static friction, so f_s supplies the net inward force. Vertical: $n = mg = 11\,760\text{ N}$.

(a) $f_s = \frac{mv^2}{R} = \frac{1200(15.0)^2}{50.0} = 5400\text{ N}$; $\mu_{s,\min} = \frac{f_s}{n} = 0.46$.

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Dry: $\sqrt{0.80 \times 9.8 \times 50.0} = 19.8\text{ m/s}$. Wet: $\sqrt{0.30 \times 9.8 \times 50.0} = 12.1\text{ m/s}$.

Evaluate: 15.0 m/s is safe when dry and unsafe when wet, which matches everyday experience. ✓

Misconception: “centrifugal force”

In the car, you feel thrown outward.

Is there an outward force pushing you?

In the ground (inertial) frame, **no outward force acts on you**. Your body tends to move in a straight line (first law), while the car turns toward the centre.

What actually happens

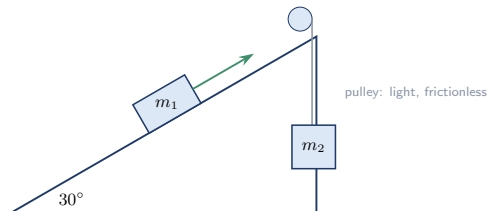
The seat and door exert a force on you toward the centre. That inward net force is what makes you follow the curve; the “outward push” is your inertia.

Integrated problem: a crate, a slope and a pulley

Decide what physics applies; do not look for a formula

A 10.0 kg crate rests on a rough 30° incline ($\mu_s = 0.30$, $\mu_k = 0.25$). A light rope runs from the crate, parallel to the slope, over a frictionless pulley to a 8.0 kg hanging block. The system is released from rest.

Does the system move? If so, in which direction? Find the acceleration and the tension.



Predict first

IDENTIFY $\rightarrow$ FBDs and axes $\rightarrow$
PREDICT $\rightarrow$ EXECUTE $\rightarrow$
EVALUATE.

Identify, set up: two bodies, connected objects (Toolkit 5), a slope (Toolkit 2) and friction (Toolkit 6). Axes: along the slope for the crate (positive up the slope), vertical for the block (positive down), consistent with the constraint.

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Does it move? Net pull toward the block: $m_2g - m_1g \sin 30^\circ = 78.4 - 49.0 = 29.4 \text{ N}$.
Maximum static friction: $\mu_s n = 0.30 \times (10.0)(9.8) \cos 30^\circ = 25.5 \text{ N}$. Since $29.4 > 25.5$, it moves, with the crate going **up** the slope.

Identify, set up: two bodies, connected objects (Toolkit 5), a slope (Toolkit 2) and friction (Toolkit 6). Axes: along the slope for the crate (positive up the slope), vertical for the block (positive down), consistent with the constraint.

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Execute: $f_k = 0.25 \times 84.9 = 21.2 \text{ N}$ (down the slope). Crate: $T - 49.0 - 21.2 = 10.0 a$.
Block: $78.4 - T = 8.0 a$.

Add: $8.2 = 18.0 a \Rightarrow a = 0.46 \text{ m/s}^2$; $T = 78.4 - 8.0(0.46) = 74.7 \text{ N}$.

Identify, set up: two bodies, connected objects (Toolkit 5), a slope (Toolkit 2) and friction (Toolkit 6). Axes: along the slope for the crate (positive up the slope), vertical for the block (positive down), consistent with the constraint.

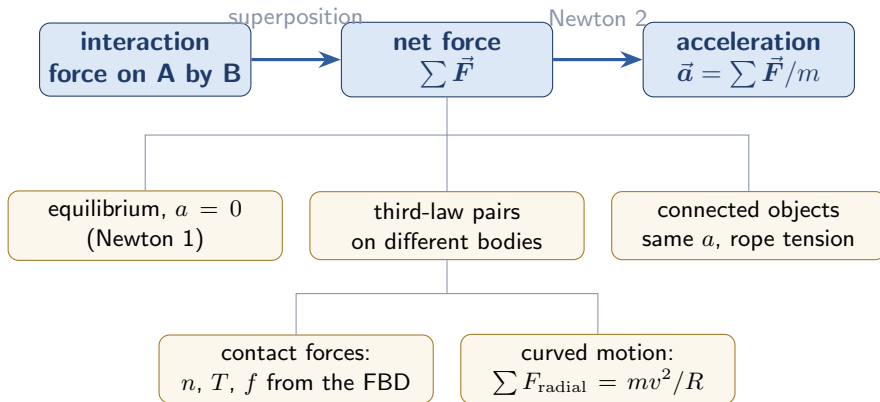
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Add: $8.2 = 18.0 a \Rightarrow a = 0.46 \text{ m/s}^2$; $T = 78.4 - 8.0(0.46) = 74.7 \text{ N}$.

Evaluate: $T < m_2g = 78.4 \text{ N}$; small a , because the pull barely exceeds the resistance;
whole-system check $m_2g - m_1g \sin 30^\circ - f_k = (m_1 + m_2)a$. ✓

Week 3 concept map



MASTER PROCEDURE – FORCE PROBLEMS

1. **IDENTIFY** Which body or bodies, and what is asked?
 2. **ISOLATE** One body at a time.
 3. **DRAW FBD(S)** Forces on the body, each named by agent (Toolkit 1).
 4. **CHOOSE AXES** Along the acceleration or the surface (Toolkit 2).
 5. **PREDICT** Signs, direction, trend, rough size.
 6. **WRITE $\sum F = ma$ BY COMPONENTS** One equation per axis per body (Toolkits 3 to 7).
 7. **SOLVE** Algebra first, then numbers with units.
 8. **EVALUATE** Units, sign, direction, limiting cases, back to the FBD.
- SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2

SET UP $\leftrightarrow$ steps 3–5

EXECUTE $\leftrightarrow$ steps 6–7

EVALUATE $\leftrightarrow$ step 8

Physical meaning

The FBD is the bridge:

situation $\rightarrow$ FBD $\rightarrow$

$$\sum F_x = ma_x, \sum F_y = ma_y$$

$\rightarrow$ motion.

THE SEVEN TOOLKITS

- | | | |
|---|---|--|
| 1 | Free-body diagrams | isolate one body; interactions; forces on the body, named by agent |
| 2 | Choosing axes and resolving forces | axes to simplify; Week 1 components; incline axes |
| 3 | Applying $\sum F = ma$ | one equation per axis; equilibrium as $a = 0$ |
| 4 | Third-law pairs | "A on B" and "B on A"; different bodies |
| 5 | Connected objects | one FBD per body; same a ; ideal rope and pulley |
| 6 | Friction and the normal force | find n ; static or kinetic; $f_s \leq \mu_s n$, $f_k = \mu_k n$ |
| 7 | Circular dynamics | radial axis; real forces only; $\sum F_{\text{radial}} = mv^2/R$ |

The Master Procedure sits above these. Week 1 and Week 2 toolkits (components, signs, equation selection) are reused, not renamed.

Net force

$$\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \cdots$$

First law

$$\sum \vec{F} = 0 \Rightarrow \vec{a} = 0$$

Second law

$$\sum F_x = ma_x, \quad \sum F_y = ma_y$$

Third law

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$$

Weight

$$w = mg$$

Friction

$$0 \leq f_s \leq \mu_s n, \quad f_k = \mu_k n$$

Incline

$$w_{\parallel} = mg \sin \theta, \quad w_{\perp} = mg \cos \theta$$

Circular

$$\sum F_{\text{radial}} = mv^2/R = m\omega^2 R$$

Six habits for Week 3

✓ Habit 1

Draw the free-body diagram before writing any equation.

✓ Habit 2

Name every force by who exerts it on whom.

✓ Habit 3

Choose axes to simplify the physics.

✓ Habit 4

Find n from the perpendicular equation; do not assume $n = mg$.

✓ Habit 5

Third-law pairs act on different bodies.

✓ Habit 6

Compare every answer with the free-body diagram and your prediction.

Checklist: can I ...

- ☐ name a force by who exerts it on whom?
- ☐ find a net force by components?
- ☐ distinguish mass, weight and normal force?
- ☐ draw a correct free-body diagram?
- ☐ apply the first and second laws by components?
- ☐ solve equilibrium problems?
- ☐ identify third-law pairs and explain why they do not cancel?
- ☐ solve connected-object problems with ideal ropes and pulleys?
- ☐ apply $f_s \leq \mu_s n$ and $f_k = \mu_k n$ with the correct direction?
- ☐ solve incline problems with sensible axes?
- ☐ explain and apply $\sum F_{\text{radial}} = mv^2/R$?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

Tutorial sheet: the lift, extra FBD, connected-object, friction, incline and circular problems, and the banked curve, conical pendulum and vertical circle.

Supplementary notes: inertial frames, the momentum form of the second law, drag and terminal speed, and the four fundamental forces.